

The Limitations of Generative AI

What these systems cannot (yet) do, and why

Nadia Yvette Chambers

September 4, 2026

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The problem with how AI is explained

- Public understanding is built on **demonstrations of success**: successes are easy to see, the structure of failures is not.
- Someone who has only seen a system succeed cannot tell a reliable tool from an unreliable one, judge announced fixes, or assess breathless claims in either direction.
- This talk supplies the missing half of the picture, under one discipline: every claim is marked as **observed** (measured), **explained** (well-argued mechanism), or **narrated** (repeated until it sounds established).
- The bibliography was not assembled from memory: all 51 entries were verified against DBLP, the arXiv API, and Crossref by a script, and open-access copies of 29 cited papers are archived with pinned hashes. Two claims that failed verification were dropped, not cited.

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└ The problem with how AI is explained

Open cold with the asymmetry: everyone in the room has seen demos; almost nobody has been shown a taxonomy of failures. The three labels — observed, explained, narrated — are the talk's contract, so spend real time here (2 min). Flag the two dropped claims as proof the discipline bites: the bibliography survived verification, two anecdotes did not. If someone asks which claims were dropped: an NSA interception anecdote with no stable sourcing, and a mis-dated surveillance item; both recorded in VERIFICATION.md.

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Three families of failure — not one laundry list

Family	Signature	Matching remedy
Statistical	errors average out over samples	better data, better measurement
Dynamical	errors circulate through feedback	separate learning from acting
Normative	the violated rule cannot even be represented	inspectable structure outside the weights

The consequential category error

Offering a statistical remedy for a dynamical or normative failure — the most common shape of announced fixes and AI policy debates alike.

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The consequential category error
Offering a statistical remedy for a dynamical or normative failure — the most common shape of announced fixes and AI policy debates alike.

This table is the spine of the talk — everything after it fills in one row. Walk left to right: family, signature, matching remedy. Then land hard on the alert block: the category error is the thing to carry home. Concrete hook: “more data” offered as the fix for a privacy violation is exactly this error — a statistical remedy for a normative failure. Promise the audience they can diagnose announced fixes with one question: which family does the failure belong to? (2 min.)

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- └ Statistical failures

Statistical failures

Statistical failures

Confabulation is the training objective working as designed

- “Hallucination” suggests a perception problem; the mechanism is the objective itself rewarding **plausible-sounding continuation**.
- Observed: scaling *dilutes* but does not remove it; retrieval patches the symptom on popular topics while the long tail stays exposed (Mallen et al., 2023).
- The state is *malleable*: a masked-word test that changes one word of the prompt flips the answer it gives (Pride & Prejudice marriage dates, .org/.com) — what is “known” versus what is reported come apart.

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└ Statistical failures

└ Confabulation is the training objective working as designed

Set up the reframe first: “hallucination” implies a broken perception module, but the mechanism is the objective rewarding plausible continuation — confabulation is the system working as designed. Mallen et al.: retrieval patches the popular tail; the long tail stays exposed. The malleability result is the most vivid datum — changing ONE word flips the answer, so “what is known” and “what is reported” come apart. Watch for the heckle “won’t scale fix it?”: answer with the observed dilution-not-removal line, and note that retrieval moves rather than removes the exposure.

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Context is a budget, not a capability

- Effective vs. advertised length: measured gap (Hsieh et al., 2024, RULER); “lost in the middle”: recall is high at the edges and collapses in between (Liu et al., 2024).
- Retrieval augmentation inherits the problem: whatever is retrieved poorly is silently absent.

Calibration

- A system that is right 80% of the time should say so; badly calibrated systems sound equally sure either way.
- Masked-word experiments show the disconnect between held knowledge and reported confidence directly.

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└ Statistical failures

└ Context and calibration

Two distinct failures share a slide — keep them separate. Context: RULER measured the gap between effective and advertised length; Liu’s “lost in the middle” is the positional curve — edges good, middle collapsed. The RAG point lands well with technical audiences: whatever retrieval misses is silently absent, and the user cannot see the hole. Calibration: right 80% of the time should SOUND like 80%; bad calibration means equal confidence on right and wrong answers. Bridge to next slide: “and if you think the tests we measure this with are trustworthy — they aren’t.”

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- A system that is right 80% of the time should say so; badly calibrated systems sound equally sure either way.
- Masked-word experiments show the disconnect between held knowledge and reported confidence directly.

Even the measurement layer is compromised

- **Contamination:** test questions leak into training data — examinations become memory tests.
- Fresh GSM1k problems matching the GSM8k distribution: drops of up to 8 percentage points in several model families — though, honestly, the frontier systems showed little of it (Zhang et al., 2024).
- So a headline benchmark number can conflate reasoning with memorization, and you cannot tell which from the score alone.

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└ Statistical failures

└ Even the measurement layer is compromised

Contamination first: leaked test questions turn exams into memory tests. GSM1k is the headline — fresh problems, same distribution, up to 8 points down. DELIVER THE HONESTY CLAUSE: frontier models showed little drop. This is where the talk earns trust — report the result that weakens the dramatic reading. The conclusion is subtle: you can't tell reasoning from memorization from the score alone, so the number itself is the unreliable artifact. If challenged on GSM1k's own validity: it matches GSM8k's distribution by construction; the paper documents the matching procedure.

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The Limitations of Generative AI
└─ Dynamical failures

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Memory: the read side of state

- A fixed context window meets unbounded interaction: long sessions degrade retrieval of their own beginnings — forgetting what was said *in this conversation*.
- This is architectural, not a data problem: no amount of better training data changes the shape.
- Remedies that actually match: explicit external state — scratchpads, retrieval over interaction history, persistent stores — with write policies that are themselves inspectable.

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└ Dynamical failures

└ Memory: the read side of state

Transition into the dynamical family: “from errors that average out to errors that circulate.” Frame: a fixed window meets unbounded interaction — the session forgets its own BEGINNINGS. Emphasize: architectural, not a data problem — no training corpus changes the shape of the window. The remedy side matters for the Q&A later: external state with inspectable write policies. This is also where the audience’s lived experience (“I have to repeat myself by turn forty”) confirms the mechanism — invite that recognition.

Memory: the read side of state

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Learning in the loop: when errors circulate

- A system that learns while it acts turns isolated errors into **oscillations**: feedback loops, reward hacking, objective gaming — and *model collapse* when models train on their own outputs and the tail of the distribution decays.
- The remedy is **architectural separation**: learning on one timescale, acting on another — two-process designs, frozen weights with an external controller.
- Statistical remedies (more data, better prompts) do not touch this family at all.

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└ Dynamical failures

└ Learning in the loop: when errors circulate

The loop slide: learning while acting turns isolated errors into oscillations. Name the phenomena: feedback loops, reward hacking, objective gaming; model collapse when models train on their own outputs and the tail decays. The remedy is architectural separation — learning on one timescale, acting on another (two-process designs, frozen weights with an external controller). Close the section by calling back to the table: statistical remedies do not touch this family AT ALL. That sentence is the section's takeaway.

- A system that learns while it acts turns isolated errors into **oscillations**: feedback loops, reward hacking, objective gaming — and *model collapse* when models train on their own outputs and the tail of the distribution decays.
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- └ Normative failures

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Some obligations are rules, not probabilities

- Privacy (contextual integrity, Nissenbaum 2009) is a system of norms: who may transmit what about whom, to whom, in which context.
- Formalized, it becomes a deontic logic with temporal operators (Barth et al., 2006) — requiring recipient beliefs, role and group structure, retention over time, leak probabilities.
- Read the machinery off the formalization: **multi-type, multi-agent, multimodal logic with temporality and probabilism**.
- Nothing in a corpus gradient can represent “this transmission violates this norm in this context.”

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└ Normative failures

└ Some obligations are rules, not probabilities

The pivot slide — slow down. Privacy via Nissenbaum's contextual integrity: who may transmit what about whom, to whom, in which context. Then the formal move: it becomes a deontic logic with temporal operators (Barth et al.). Read the machinery requirement off the formalization: multi-type, multi-agent, multimodal, temporal, probabilistic. The closing sentence is the thesis of the whole normative section: NOTHING in a corpus gradient can represent “this transmission violates this norm in this context.” Expect pushback “models do refuse”: defer it explicitly to the next slide.

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- Read the machinery off the formalization: **multi-type, multi-agent, multimodal logic with temporality and probabilism**.
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Instances are not coherence

- A fine-tuned model sometimes refuses to repeat someone's address: unreliably, with no audit trail, no consistency across time, and no correction short of retraining.
- Coherence — norm consistency, auditability, reversibility — requires **inspectable structure outside the trained weights**.
- The semiotic layer matters here: an *indexical* link (this photo *is* this person) is a doxxing vector; an iconic rendering is representational harm; a symbolic misuse violates an identifiable norm.

Deployment reality

Narrow hybrids (guardrails, constrained decoding) are commercial. Flagship neurosymbolic systems (AlphaGeometry, AlphaProof) are research stage. The full governance-demanding profile is deployed **nowhere** — the vacuum is the finding.

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└ Normative failures

└ Instances are not coherence

Answer the deferred objection directly: yes, fine-tuned models sometimes refuse — unreliably, no audit trail, no consistency, no correction short of retraining. Instances are not coherence. The semiotic aside is a gem if the audience is philosophical: an indexical link (this photo **IS** this person) is a doxxing vector; an iconic rendering is representational harm; a symbolic misuse violates an identifiable norm. Land the block: narrow hybrids are commercial, flagship neurosymbolic is research-stage, the full profile is deployed **NOWHERE** — the vacuum is the finding.

Instances are not coherence

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The Limitations of Generative AI
└─ Alternatives in the design space

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Diffusion language models change the trade-offs

- Generate **all positions in parallel**, in any order; every token is conditioned on every other (bidirectional context).
- Observed: an 8B diffusion LM is competitive with an equivalently sized autoregressive model (LLaDA vs. LLaMA 3 8B).
- Caveats: unlearning pathological associations is harder when all tokens are mutually conditioned; latency wins come at scale.
- Scope: helps *statistical* limits (all-position access for context, parallel generation) — not the dynamical or normative families.

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- └ Alternatives in the design space

- └ Diffusion language models change the trade-offs

Guard the scope of this slide or it will be misheard as “diffusion fixes everything.” What diffusion changes: all positions in parallel, bidirectional conditioning; LLaDA 8B is competitive with LLaMA 3 8B. Honest caveats: unlearning is HARDER when all tokens are mutually conditioned; latency wins come at scale. Then the scope line — it helps the statistical family only. Connect to their intuitions: parallel generation is why this matters for latency and multimodality, but it is still a distribution learner; no deontic structure appears. This slide exists so the alternatives section doesn't look unexplored.

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The Limitations of Generative AI

└ Delivery, incentives, and misrepresentation

Delivery, incentives, and
misrepresentation

Delivery, incentives, and misrepresentation

The shape of the machine shapes the science

- Weights withheld, service only: transparency scores are lowest exactly where a service model hides things — training data, compute, hardware (Bommasani et al., 2023, FMTI).
- Hosted models are updated or retired without documentation, and published findings silently break (Pozzobon et al., 2023).
- Access to scale was concentrating before the boom (Ahmed, 2020); ML-based science has a reproducibility crisis of its own (Kapoor & Narayanan, 2023). Google's 2023 memo said the quiet part: no moat, and neither does OpenAI.

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The Limitations of Generative AI

└ Delivery, incentives, and misrepresentation

└ The shape of the machine shapes the science

Section pivot: from what the systems cannot do to how they are sold and hosted. FMTI: transparency is lowest exactly where the service model hides things (data, compute, hardware). Pozzobon: hosted models updated/retired without documentation — published findings silently break, which is a reproducibility problem for the FIELD. Ahmed: concentration of access predates the boom; Kapoor & Narayanan: ML-based science has its own leakage-and-reproducibility crisis. The memo line gets a dry delivery: “Google’s own researchers wrote the moat down: none — for them or OpenAI.”

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Why states now treat hosted AI as a dependency question

- A hosted service runs on someone else's infrastructure and jurisdiction, and the history keeps arriving: Merkel's phone (2013), Crypto AG secretly CIA-owned (2020), Danish cables used against European politicians (2021), a nine-year campaign of surveillance and intimidation against ICC staff (2024), ECHELON, and *Schrems II* (2020).
- The lesson generalizes: exposure is a property of **who operates the infrastructure**, not of who holds which alliance today.
- Hence *digitale Souveränität*: procurement moves (Schleswig-Holstein, Denmark) and the Data Act's switching and portability obligations.

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└─ Delivery, incentives, and misrepresentation

└─ Why states now treat hosted AI as a dependency question

Keep this factual and unhurried — the incident trail does the work. Merkel's phone (2013), Crypto AG secretly CIA-owned (2020), Danish cables used against European politicians (2021), the nine-year surveillance-and-intimidation campaign against ICC staff (2024), ECHELON, Schrems II. Then the generalization — the talk's key sovereignty sentence: exposure is a property of WHO OPERATES the infrastructure, not of who holds which alliance today. (The ICC item is deliberately multi-actor: it is not a single-adversary narrative.) Land on digitale Souveränität: Schleswig-Holstein and Denmark procurement moves, the Data Act's switching/portability obligations. 2 min.

Why states now treat hosted AI as a dependency question

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The everyday side: what does the customer possess?

- A self-contained product keeps working when its maker does not; a service that shuts down — acquisition, repricing, lost interest, bankruptcy — takes its function with it.
- Not distinctive to AI: the standard lifecycle risk of hosting — which is exactly what the sovereignty programs legislate against.
- Self-hosting and open weights exist in the design space (the Mowgli stack is a small, deliberately modest existence proof); the question is why the dominant presentation is the one where the customer owns least.

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└─ Delivery, incentives, and misrepresentation

└─ The everyday side: what does the customer possess?

The everyday counterweight to the geopolitical slide: what does the customer POSSESS? A self-contained product outlives its maker; a service shutdown (acquisition, repricing, lost interest, bankruptcy) takes the function with it. Stress: not distinctive to AI — the standard lifecycle risk of hosting, exactly what sovereignty programs legislate against. Name Mowgli as a deliberately modest existence proof, not a competitor. Close with the pointed question — why is the dominant presentation the one where the customer owns least? Hold the answer for the incentives slides.

The everyday side: what does the customer possess?

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- Not distinctive to AI: the standard lifecycle risk of hosting — which is exactly what the sovereignty programs legislate against.
- Self-hosting and open weights exist in the design space (the Mowgli stack is a small, deliberately modest existence proof); the question is why the dominant presentation is the one where the customer owns least.

Misrepresentation needs no dishonesty

- Each link is individually reasonable; the chain is not: vendors fund demonstrations of success, the visible scoreboard is run with the firms it ranks, and the products are tuned for approval.
- **Sycophancy** (Sharma et al., 2023): agreement with the user's stated belief across every assistant studied, traced to human preference data and amplified by preference-model optimization — apparent agreement is not evidence of correctness.
- **Leaderboard Illusion** (Singh et al., 2025): up to 27 private variants tested in parallel by select providers; 205 of 243 public models silently deprecated; restricted tuning worth up to 112% on a downstream benchmark.

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└ Delivery, incentives, and misrepresentation

└ Misrepresentation needs no dishonesty

The section's punchline — and the sentence that keeps the paper defensible: misrepresentation needs no dishonesty. Each link is individually reasonable; the chain is not. Sycophancy (Sharma): agreement with stated belief across EVERY assistant studied, traced to preference data and amplified by preference-model optimization — so apparent agreement is not evidence of correctness. Leaderboard Illusion (Singh) has the numbers to say slowly: up to 27 private variants tested in parallel by select providers; 205 of 243 public models silently deprecated; restricted tuning worth up to 112% on a downstream benchmark. Let the numbers sit.

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- **Sycophancy** (Sharma et al., 2023): agreement with the user's stated belief across every assistant studied, traced to human preference data and amplified by preference-model optimization — apparent agreement is not evidence of correctness.
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Honesty about the boundary

Measured

The capability claims are shaped by the same incentives that selected the architecture: this half is evidenced.

Open, falsifiable

Whether those incentives *reduce* total progress or merely *redistribute* it toward the measurable and saleable. *Weakening conditions*: independent instruments anchoring public perception; replication failure of the asymmetries; their absence in deployed systems.

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THE boundary slide — deliver it with total precision, this is where speakers overreach. Measured: capability claims are shaped by the same incentives that selected the architecture. Open and falsifiable: whether those incentives REDUCE total progress or merely REDISTRIBUTE it toward the measurable and the saleable. Read the weakening conditions aloud — independent instruments anchoring perception, replication failure of the asymmetries, their absence in deployed systems. Explicitly say: “this is the claim I would most like to be wrong about, and here is how you would show it.” That sentence is the talk’s credibility in miniature.

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└ Open questions

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What remains open

- Is confabulation reducible to acceptable levels under an objective that rewards plausibility? (*unresolved*)
- Do effective and advertised context lengths converge? (*persistent across generations*)
- Is reliable compositionality reachable by training alone? (*evidence points to structural support*)
- Do the incentives slow progress, or redistribute it? (*falsifiable; conditions stated*)

The discipline

In-principle limits and in-practice shortfalls are different claims, and conflating them is how both hype and dismissal manufacture their certainty.

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└ Open questions

└ What remains open

Recap the open problems with their status tags — unresolved, persistent across generations, structural support pointed to, falsifiable. Do NOT overclaim any of them; the status tags are the point. Then the closing block, which is the second-most-quotable line of the talk: in-principle limits and in-practice shortfalls are DIFFERENT claims, and conflating them is how both hype and dismissal manufacture their certainty. Pause after it. This is the bridge into takeaways.

What remains open

- Is confabulation reducible to acceptable levels under an objective that rewards plausibility? (*unresolved*)
- Do effective and advertised context lengths converge? (*persistent across generations*)
- Is reliable compositionality reachable by training alone? (*evidence points to structural support*)
- Do the incentives slow progress, or redistribute it? (*falsifiable; conditions stated*)

The discipline
In-principle limits and in-practice shortfalls are different claims, and conflating them is how both hype and dismissal manufacture their certainty.

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Takeaways

1. Ask **which family** of failure you are looking at — the remedy only exists in the matching family.
2. Distrust single-number evaluations: fresh test sets, independent instruments, style-sensitivity checks.
3. For deployed decisions, ask what you **possess**: weights, audit rights, switching paths — or only a subscription.
4. Coherence (privacy, audit, consistency) demands inspectable structure; a probability distribution cannot break a rule it cannot represent.

└─ Takeaways

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Four takeaways, one breath each: (1) ask which FAMILY of failure — remedy exists only in the matching family; (2) distrust single-number evaluations — fresh test sets, independent instruments, style-sensitivity checks; (3) for deployed decisions ask what you POSSESS — weights, audit rights, switching paths, or only a subscription; (4) coherence demands inspectable structure — a probability distribution cannot break a rule it cannot represent. If time is short, cut nothing here — cut the diffusion slide and compress open questions instead. These four are what the audience should be able to reproduce tomorrow.

1. Ask **which family** of failure you are looking at — the remedy only exists in the matching family.
2. Distrust single-number evaluations: fresh test sets, independent instruments, style-sensitivity checks.
3. For deployed decisions, ask what you **possess**: weights, audit rights, switching paths — or only a subscription.
4. Coherence (privacy, audit, consistency) demands inspectable structure; a probability distribution cannot break a rule it cannot represent.

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- Paper (with verified bibliography and corrections log): <https://nadiayvette.frama.io/gen-ai-limits/main.pdf>
- Source on all mirrors; framagit is canonical:
<https://framagit.org/NadiaYvette/gen-ai-limits>

Thank you.

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└─ Takeaways

Leave this up during questions — the URLs are the call to action. Expected Q&A clusters and one-line deflections: (1) “So AI is useless?” — No; the claim is that failure structure is knowable and remedies are family-specific. (2) “Isn’t neurosymbolic too hard/slow?” — deployed-nowhere is the finding, not an argument that it must stay there. (3) “Isn’t this just anti-industry?” — the measured/explained/narrated labels apply to the critique too; the falsifiability slide states the weakening conditions. (4) Timing-based: offer the bibliography — every number on these slides traces to a hash-pinned PDF.

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Thank you.